

Master Project Blueprint: AI-Driven Compliance & Legal Auditor

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Project: AI-Driven Compliance & Legal Auditor

Technology Stack: MERN + Gemini 1.5 Pro (Large Context LLM) + PDF.js

Delivery Timeline: 4-Week Rapid Sprint

1. Executive Summary & Product Vision

Manual legal document review is notoriously slow, expensive, and prone to human error, exposing enterprises to massive financial and regulatory risks.

The **AI-Driven Compliance & Legal Auditor** is a specialized enterprise tool designed to automate the initial review of contracts, NDAs, and employee handbooks. By leveraging Large Language Models with massive context windows (like Gemini 1.5 Pro), the platform can ingest entire 50-page contracts in seconds, cross-reference them against a company's internal compliance playbook or regional labor laws, and visually "Red Flag" risky clauses. This is not designed to replace lawyers, but to act as a highly efficient paralegal that drastically reduces billable hours and ensures zero critical oversights.

2. End-to-End Feature List

Module 1: Secure Document Ingestion & Parsing

- **Multi-Format Upload:** Secure drag-and-drop ingestion for .pdf, .docx, and .txt files.
- **Text Extraction Engine:** Backend processing using pdf-parse or mammoth.js to strip raw text while preserving clause numbering and paragraph structures.
- **Document Vault:** A highly secure, encrypted repository of all uploaded legal documents, sorted by client, contract type, and date.

Module 2: AI Risk Detection & Clause Analysis

- **Automated Red-Flagging:** The AI scans the document and highlights clauses that represent a high risk (e.g., infinite liability, missing arbitration clauses, unfavorable termination rights).
- **Compliance Cross-Referencing:** The AI compares the uploaded document against the company's "Standard Playbook" (e.g., "Our company never accepts Net-90 payment terms; flag if present").

- **Plain-English Summarization:** The AI generates a 1-page executive summary of a 50-page contract, explaining the key obligations and liabilities in simple terms for business executives.

Module 3: Version Control & Smart Diffing

- **Contract Comparison (Diffing):** Upload V1 and V2 of a contract. The AI doesn't just show what words changed; it explains the *legal implication* of the change (e.g., "The client changed the termination notice from 30 days to 15 days").
- **Version History:** Track the evolution of a document through multiple rounds of negotiation.

Module 4: Interactive Audit Dashboard

- **Split-Screen Viewer:** A React-based document viewer showing the original PDF on the left and the AI-generated risk audit on the right.
- **Interactive Annotations:** Users can click an AI-flagged risk, read the AI's explanation, and click "Accept Risk", "Reject", or "Draft Counter-Clause".
- **AI Counter-Drafting:** If a clause is flagged as risky, the AI can instantly generate a revised, legally protective version of that specific clause for the user to copy-paste.

Module 5: System Administration & Export

- **Audit Report Generation:** One-click export of a formal PDF report detailing all found risks, ready to be sent to the external legal counsel.
- **Role-Based Access Control (RBAC):** Strict permissions ensuring that only authorized executives or legal personnel can view highly confidential contracts.

3. System Architecture & Data Flow

The core Audit Workflow follows this strict linear pipeline:

1. **React Frontend:** User uploads a contract (e.g., "Vendor_Agreement_v1.pdf").
2. **Node/Express Backend:** Receives file, strips text, and temporarily stores it in memory.
3. **Google Gemini API (1.5 Pro):** The backend sends the entire text + the "Company Playbook Rules" to the LLM with a strict prompt: *"Identify risks, output exact quote, and provide risk severity (Low, Med, High) in JSON."*
4. **Backend Logic:** Formats the JSON into an **AuditReport** object.
5. **MongoDB:** Saves the document metadata and the attached Audit Report.
6. **React Frontend:** Renders the Split-Screen Viewer, mapping the JSON risk coordinates to the text to highlight them in red/yellow visually.

4. Database Schema (MongoDB Phase 1)

Users Collection

Field Name	Data Type	Description
_id	ObjectId	Unique identifier
name	String	Full name of the employee/lawyer
email	String	Corporate email
department	String	E.g., 'HR', 'Legal', 'Procurement'

Playbooks Collection (Compliance Rules)

Field Name	Data Type	Description
_id	ObjectId	Unique rule identifier
ruleName	String	E.g., "Maximum Liability Cap"
ruleDescription	String	E.g., "Never accept liability over \$1M."
isActive	Boolean	Toggle for the AI to use this rule

Documents Collection

Field Name	Data Type	Description
_id	ObjectId	Unique document identifier
fileName	String	E.g., "Service_Contract.pdf"
uploadDate	Date	Timestamp of ingestion
status	String	Enum: ['Pending', 'Audited', 'Approved']
fileUrl	String	Secure S3/Cloud Storage link

AuditReports Collection

Field Name	Data Type	Description
_id	ObjectId	Unique audit identifier
documentId	ObjectId	Reference to Documents collection
executiveSummary	String	AI-generated overview
flaggedClauses	Array	Array of { quote, issue, severity, suggestedFix }
reviewedBy	ObjectId	User ID of the human who approved the audit

5. Four-Week Sprint Plan (Developer Roadmap)

Timeline	Frontend (React/Tailwind)	Backend (Node/Express/MongoDB)	QA & Testing
Week 1: Document Vault	Setup UI framework, Auth Screens, and the Document Upload/List views.	Setup DB Schema, JWT Auth, integrate file upload (Multer) & Text Extraction (pdf-parse).	Test text extraction accuracy on complex, multi-column PDFs.
Week 2: The AI Auditor	Build the Split-Screen Viewer UI and Risk Card components.	Integrate Gemini API. Develop the strict prompt engineering for legal analysis.	Feed known "bad" contracts to the AI; verify it catches the intentional errors.
Week 3: Interaction & Rules	Build the Playbook Management UI (where admins add rules) and AI Counter-Drafting buttons.	Build endpoints to save/update Audit Reports and connect Playbook rules to the AI prompt.	Test Context Window limits (feed a massive 100-page document to ensure it doesn't crash).
Week 4: Export & Security	Build the PDF Export feature for the final Audit Report.	Security hardening, API rate limiting, ensure temporary text files are deleted from the server.	End-to-end security penetration testing (critical

			for legal software).
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6. Non-Functional Requirements & Success Criteria

- **Security & Confidentiality (Absolute Priority):** Legal documents are highly sensitive. No data can be used to train public AI models (ensure Enterprise API tiers are used). Documents must be encrypted at rest in the database/storage.
- **Context Window Handling:** The system must flawlessly handle documents up to 50,000 words without timing out or losing context.
- **Client Hand-off Goal:** By the end of Week 4, the team must demonstrate uploading a complex 20-page Non-Disclosure Agreement (NDA) and having the system instantly flag a hidden, unfavorable "non-compete" clause, complete with a suggested legal rewrite.

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